Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Daniel Bosma	dbosma@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

My project will be a Tesla Vehicle Simulator. The goal is to allow users to learn about and test Tesla vehicles.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

There is currently a tesla simulator mobile game app that is available to purchase on Google Play. There is also a Tesla self-driving VR game on Roblox.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

In this program, users will be able to simulate driving Tesla Vehicles. Users will select a desired starting balance and pick a Tesla vehicle to purchase. Depending on the money in their account, the user can race their vehicle or navigate to different locations. If their car gets in a crash, users can improve their car's functionality by paying for repairs (or simply buying extra features without crashing) in the simulator shop. If their vehicle's battery is low

and a destination can't be reached, users will need to charge their car. Users must watch their battery level to avoid running out of range and getting towed! After the user is finished driving, they can view their trip stats, which include the places they traveled, the total money spent, the miles driven, total time driving, gas money saved, and a recommendation from the website on whether they should purchase the vehicle in real life or not. My website will allow users to get a good idea of what it is like to drive a Tesla vehicle.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

My program will estimate the total cost to drive a Tesla in real life vehicle based on the car a user chooses, where the user chooses to go, any accidents they get in, and Tesla charging prices, and furthermore will either recommend or not recommend that the user purchases the vehicle in real life.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

The user will need to enter a starting balance, purchase a desired vehicle, enter weather conditions, and choose locations to go to. They will also need to decide when they want to charge their car, and what upgrades or repairs (if they crash) to implement.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

- Based on a user's starting balance, the computer will filter out vehicle options that the user can't afford.
- When choosing a location to travel to, the computer will decide which locations the user should travel to by calculating if their remaining account balance is greater than or equal to the cost to attend each location and by calculating if their remaining range is greater than or equal to the distance to each location.
- If the user sets the weather to rainy, the computer will double any new driving time to the user's total driving time.
- When charging, the computer will add range to the vehicle and take money out of the user's account.

- The computer will calculate how much money the user saved on gasoline. To do this, it will first calculate the cost to drive the selected car by dividing miles driven by the MPGE (miles-per-gallon equivalent) of their selected car and then multiplying by the price of gas. Then, it will find the cost to drive a gas car of the same size and weight over the same miles by dividing miles driven by the MPG of the same size and weight gas car and then multiplying by the price of gas.
- When the user decides to stop using the simulation, the program will calculate a decision on whether the user should or shouldn't purchase the vehicle they tested in real life.
    - If the user has driven to 5 locations, has saved $50 from not using gas, and has raced their vehicle at least once, the decision will be ‘yes’. Otherwise, the decision will be ‘no’.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

- When selecting a vehicle, the user will be presented vehicles that only they can afford
- When choosing a location to travel to, the user
- will see a 2D list of locations to travel to, upon which they can click.
    - “Charging needed” will be displayed on locations that the user doesn't have enough range to get to.
    - “Insuficent balance” will be displayed on locations that the user can't afford to attend.
- After driving and attending any location, the user’s simulation stats (including locations visited, money spent, remaining balance, money saved on gas, vehicle battery percentage, and vehicle’s MPGE will all be shown
- If user crashes, they will see an updated screen with the decreased MPGE of their vehicle
- If the user selects a location that is too far away, a screen will appear saying that their car ran out of range and was towed, along with their updated account balance after paying for the tow.
- When the user is finished driving, they will see their final stats and a recommendation on whether or not they should purchase the car they tested in real life.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

- The User's Selected Car
- The user's account
- The list of locations the user can travel to
- The current weather condition

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

For the users selected car, I need to store variables for it's:
- Max Range
- Current Battery Percentage
- Price
- MPGE (Miles Per Gallon Equivalent)

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

- *The math: The MPGE and the distance traveled to a location will need to be used to calculate the remaining miles after arriving at a destination*
- *Displaying list of locations the user can travel to and where the user's car currently is located*

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I am going to code the classes, variables, and non-route functions needed in my project and then have an LLM (potentially ChatGPT) code the remaining drafter functions.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My ultimate goal is to prove to myself that I can make a cool and functional program, and to learn more about electric vehicles in the process.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to build a simple web tool to help people understand more about electric vehicles and how to use them. What are 3 common misconceptions people have about electric vehicles?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

I should consider implementing:
1. Cost Savings Calculator: Calculates how much money users can save by driving an electric vehicle instead of a gasoline-powered car.

2. Range Calculator: Estimates how many miles an electric vehicle can travel based on its battery percentage and maximum driving range.

3. Charging Time Calculator: Estimates how long it will take to charge an electric vehicle based on its battery capacity, current charge level, desired charge level, and charger power.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

I think the range calculator is a little overambitious. Could you try simplifying the amount of variables needed for the formula for this?

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

ChatGPT suggested simplifying the range calculator equation to Remaining Range = Maximum Range x Battery% / 100. This convinces me that most of the math needed in my project can probably also be simplified so it is less challenging to code.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I will use the CISC help hours google calendar to find an office hours session to attend and I will ask more questions to ChatGPT. Also, I will take time away from the project to clear my mind so I can focus better.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

- *Start from the beginning of the program* - *Click through the program until the problem is encountered.* - *Attempt to fix the issue* - *Relaunch the program and see if the issue was resolved; If not, try to fix again and check again*

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously,

these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

The computer will calculate how much money the user saved on gasoline by dividing the miles the user drove by the MPGE (miles-per-gallon-equivalent) of their selected car and then multiplying by the price of gas per gallon.

Your sentence is grammatically correct, but there's an important mathematical distinction: your formula calculates the gasoline-equivalent cost of driving, not necessarily the money saved.

Your formula

$$\text{Cost} = \frac{\text{Miles Driven}}{\text{MPGe}} \times \text{Gas Price}$$

To calculate the actual money saved by driving an electric vehicle, you would need to subtract the cost of electricity from the gasoline cost of a comparable gas-powered vehicle.

Suggested revision

The computer will estimate the equivalent cost of gasoline by dividing the miles driven by the MPGe (miles-per-gallon-equivalent) of the user's selected car and multiplying the result by the price of gasoline per gallon.

Important: MPGe measures energy efficiency, not actual gasoline consumption. For a more accurate savings calculation, you should use the MPG of a comparable gasoline-powered vehicle and subtract the electric vehicle's charging cost.

I want to build a simple web tool to help people understand more about electric vehicles and how to use them. What are 3 common misconceptions people have about electric vehicles?

Here are 3 common misconceptions about electric vehicles (EVs) that would be great to address in your website project.

1. "Electric vehicles are more expensive than gas-powered cars."

Misconception: EVs always cost more to own and operate than gasoline vehicles.

Reality: Although some EVs have higher upfront prices, they often cost less to operate because electricity is generally cheaper than gasoline per mile. EVs also typically require less routine maintenance.

Website feature: A Cost Savings Calculator that compares the cost of driving an EV with a gasoline-powered vehicle based on miles driven, electricity prices, and gasoline prices.

2. "Electric vehicles can't travel very far."

Misconception: EVs don't have enough range for everyday driving or long trips.

Reality: Many modern EVs can travel 250–350 miles on a full charge, although range varies by model, weather, and driving conditions.

Website feature: A Range Calculator that estimates how far a selected EV can travel based on its battery percentage and estimated full-charge range.

3. "Charging an electric vehicle takes too long."

Misconception: Charging an EV always takes several hours, making it inconvenient.

Reality: Charging time depends on the charger and vehicle. DC fast chargers can add substantial range in approximately 20–40 minutes, while home charging typically takes several hours and can happen overnight.

Website feature: A Charging Time Calculator that estimates how long an EV will take to charge based on battery capacity, current charge level, desired charge level, and charger power.

For your Python web development project, I would recommend incorporating all three calculators into one website. They address common misconceptions while giving users interactive tools to learn about EV ownership.

Summarize the 3 website features you suggested.

1. Cost Savings Calculator: Calculates how much money users can save by driving an electric vehicle instead of a gasoline-powered car.

2. Range Calculator: Estimates how many miles an electric vehicle can travel based on its battery percentage and maximum driving range.

3. Charging Time Calculator: Estimates how long it will take to charge an electric vehicle based on its battery capacity, current charge level, desired charge level, and charger power.

I think the range calculator is a little overambitious. Could you try simplifying the amount of variables needed for the formula for this?

You can simplify the Range Calculator to use just two variables:

1. Maximum Range (miles): How far the selected EV can travel on a full charge.
2. Current Battery Percentage (%): How much battery the vehicle currently has.

Formula

$$\text{Remaining Range} = \text{Maximum Range} \times \frac{\text{Battery \%}}{100}$$

Example

Suppose an EV has a maximum range of 300 miles and currently has 60% battery.

$$300 \times \frac{60}{100} = 180 \text{ miles}$$

Simplified website feature

Range Calculator: Estimates how many miles an electric vehicle can travel by multiplying its maximum driving range by its current battery percentage.

This would be straightforward to implement in Python and would only require two user inputs. It's an estimate, since actual range depends on driving conditions, temperature, and speed.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.